

Chapter 5: Bessel Functions

Mathematical Physics

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Outline

- 1 Generating function
- 2 Recursion formulae
- 3 Integral representation
- 4 Asymptotics
- 5 Neumann functions
- 6 Hankel functions and Fourier–Bessel series
- 7 Summary

Where we are going

Bessel's equation of integer order m is

$$x^2 y''(x) + xy'(x) + (x^2 - m^2)y(x) = 0. \quad (5.0.1)$$

It arises when variables are separated in circular or cylindrical geometries. This chapter develops

- a generating function for the integer-order family J_m , and the power series;
- the order and derivative recurrences;
- one carefully derived real integral representation (plus a brief contour form);
- the small- and large-argument expansions (large x by stationary phase, the real oscillatory case of steepest descent from Ch. 1);
- the second solution Y_m , a short look at Hankel functions $H^{(1)}, H^{(2)}$, and the discrete Fourier–Bessel series on the disk.

Throughout the generating-function sections, $m, n \in \mathbb{Z}$ unless stated otherwise. For small- or large-argument limits, the order is held fixed.

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Generating function for the family $\{J_m\}$

Why a generating function? Circular problems give one radial function for each integer angular order m . Package the functions into one Laurent series:

$$g(x, t) = \exp\left[\frac{x}{2}(t - t^{-1})\right] = \sum_{m=-\infty}^{\infty} J_m(x)t^m, \quad t \neq 0. \quad (5.1.1)$$

The notation $[t^m]g$ means the coefficient of t^m , so $J_m(x) = [t^m]g(x, t)$. On $t = e^{i\theta}$, the exponential becomes $e^{ix \sin \theta}$: this explains its connection to waves in circular geometry. Recall Laurent coefficient extraction from Chapter 1:

$$[t^m]f(t) = \frac{1}{2\pi i} \oint_C f(t)t^{-m-1} dt, \quad (5.1.2)$$

where C is a counterclockwise circle about zero inside the annulus of convergence. Only the term proportional to t^{-1} survives the integration.

We will multiply the two exponential series to recover the Chapter 3 Bessel series, then differentiate the same generating function to relate neighboring orders.

Reading off the integer-order series

Expand, then collect one power. Since $g = e^{xt/2} e^{-x/(2t)}$,

$$e^{xt/2} = \sum_{j=0}^{\infty} \frac{(x/2)^j}{j!} t^j, \quad e^{-x/(2t)} = \sum_{k=0}^{\infty} \frac{(-1)^k (x/2)^k}{k!} t^{-k}.$$

Their product is

$$g(x, t) = \sum_{j,k \geq 0} \frac{(-1)^k}{j!k!} \left(\frac{x}{2}\right)^{j+k} t^{j-k}.$$

For $t \neq 0$ both exponential series converge absolutely, so we may group terms. To obtain t^m with $m \geq 0$, require

$$j - k = m \implies j = k + m, \quad j + k = 2k + m.$$

Substitute this into the coefficient of the double sum:

$$J_m(x) = [t^m]g(x, t) = \sum_{k=0}^{\infty} \frac{(-1)^k}{k!(m+k)!} \left(\frac{x}{2}\right)^{2k+m}, \quad m = 0, 1, 2, \dots \quad (5.1.3)$$

Reading off the integer-order series (cont.)

For example,

$$J_0(x) = 1 - \frac{x^2}{4} + \frac{x^4}{64} - \frac{x^6}{2304} + \cdots,$$

$$J_1(x) = \frac{x}{2} - \frac{x^3}{16} + \frac{x^5}{384} + \cdots,$$

$$J_2(x) = \frac{x^2}{8} - \frac{x^4}{96} + \frac{x^6}{3072} + \cdots.$$

These are the series obtained from the differential equation in Chapter 3. A single generating function now holds all the integer orders at once. The general-order function J_ν below is not a Laurent coefficient when $\nu \notin \mathbb{Z}$.

General order and direct verification of Bessel's equation

Replacing factorials by the gamma function extends the series to general order:

$$J_\nu(x) = \sum_{k=0}^{\infty} \frac{(-1)^k}{k! \Gamma(k + \nu + 1)} \left(\frac{x}{2}\right)^{2k+\nu}. \quad (5.1.4)$$

At a nonpositive integer, $1/\Gamma$ is defined by continuity and equals zero. Use this formula directly at negative integer order; do not divide a recurrence by zero. For $x > 0$, $(x/2)^\nu = e^{\nu \log(x/2)}$ is unambiguous; complex x requires a fixed logarithm branch when ν is noninteger.

For fixed order, the ratio of consecutive nonzero terms eventually has magnitude $|x|^2/[4(k+1)(k+\nu+1)] \rightarrow 0$. The power-series factor converges for every finite x . For $\nu = m \geq 0$, $\Gamma(k+m+1) = (k+m)!$, recovering (5.1.3).

General order and direct verification of Bessel's equation (cont.)

The same coefficients as in Frobenius. For $m \geq 0$, write $J_m = \sum_{k \geq 0} a_k x^{2k+m}$. Chapter 3's calculation makes the coefficient of x^{2k+m} in Bessel's equation

$$4k(k+m)a_k + a_{k-1}, \quad k \geq 1.$$

For the coefficients just obtained,

$$\frac{a_k}{a_{k-1}} = -\frac{1}{4k(k+m)},$$

so every coefficient cancels. The lowest-power coefficient is already zero, giving

$$x^2 J_m'' + x J_m' + (x^2 - m^2) J_m = 0. \quad (5.1.5)$$

The gamma recurrence gives the same cancellation for ν in place of m , including vanishing coefficients when interpreted through $1/\Gamma$.

Negative integer orders: $J_{-m} = (-1)^m J_m$

In the double sum for g , a coefficient of t^{-m} requires $j - k = -m$. For $m \geq 0$, this means $k = j + m$. Thus

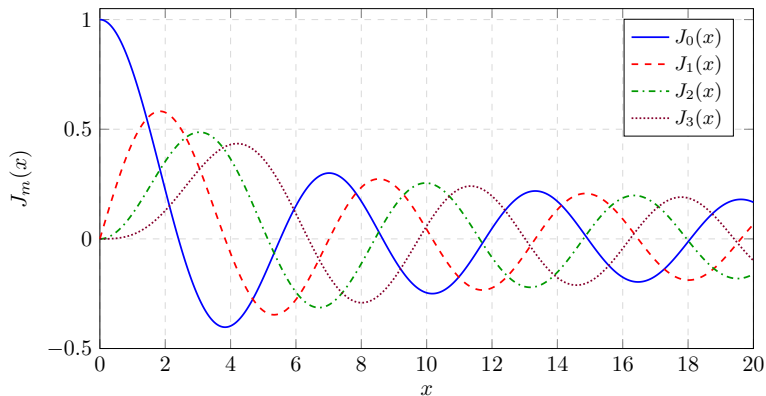
$$\begin{aligned} J_{-m}(x) &= \sum_{j=0}^{\infty} \frac{(-1)^{j+m}}{j!(j+m)!} \left(\frac{x}{2}\right)^{2j+m} \\ &= (-1)^m \sum_{j=0}^{\infty} \frac{(-1)^j}{j!(j+m)!} \left(\frac{x}{2}\right)^{2j+m}. \end{aligned}$$

The remaining sum is J_m , so

$$J_{-m}(x) = (-1)^m J_m(x), \quad m \in \mathbb{Z}. \quad (5.1.6)$$

The gamma series agrees: its first m coefficients at order $-m$ vanish because $1/\Gamma$ is zero at nonpositive integers. In particular, $J_{-1} = -J_1$ and $J_{-2} = J_2$. Negative integer orders therefore do not provide a second independent solution.

Bessel functions of the first kind



For $m \geq 0$, the leading term in (5.1.3) is $J_m(x) = (x/2)^m/m! + O(x^{m+2})$. Thus $J_0(0) = 1$ and $J_m(0) = 0$ for $m \geq 1$.

Jacobi–Anger: the generating function on $|t| = 1$

Set $t = e^{i\theta}$ in (5.1.1). Since $t - t^{-1} = 2i \sin \theta$ and $t^m = e^{im\theta}$,

$$e^{ix \sin \theta} = \sum_{m=-\infty}^{\infty} J_m(x) e^{im\theta}. \quad (5.1.7)$$

This is the Jacobi–Anger expansion: the Bessel functions are Fourier coefficients of a plane wave written in angular variables.

To isolate J_n , multiply by $e^{-in\theta}$ and integrate over one period. For an integer q ,

$$\int_0^{2\pi} e^{iq\theta} d\theta = \begin{cases} 2\pi, & q = 0, \\ (e^{2\pi iq} - 1)/(iq) = 0, & q \neq 0. \end{cases}$$

The series may be integrated term by term on this circle. All terms with $m \neq n$ vanish, leaving

$$J_n(x) = \frac{1}{2\pi} \int_0^{2\pi} e^{i(x \sin \theta - n\theta)} d\theta, \quad n \in \mathbb{Z}. \quad (5.1.8)$$

We will convert this to a real cosine integral after deriving the recurrences.

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Differentiating in t : the three-term recurrence

Differentiate $g = e^{x(t-t^{-1})/2}$ and multiply by t before comparing coefficients:

$$tg_t = \frac{x}{2}(t + t^{-1})g, \quad tg_t = \sum_{m=-\infty}^{\infty} mJ_m(x)t^m. \quad (5.2.1)$$

Multiplication by t moves the coefficient of t^{m-1} to t^m ; multiplication by t^{-1} moves the coefficient of t^{m+1} to t^m . Consequently,

$$[t^m](tg) = J_{m-1}(x), \quad [t^m](t^{-1}g) = J_{m+1}(x).$$

Matching the coefficient of t^m in (5.2.1) gives

$$mJ_m(x) = \frac{x}{2} [J_{m-1}(x) + J_{m+1}(x)].$$

For $x \neq 0$, divide by $x/2$:

$$J_{m-1}(x) + J_{m+1}(x) = \frac{2m}{x} J_m(x). \quad (5.2.2)$$

At $x = 0$, use limits from the power series, not direct substitution into a quotient.

Differentiating in x : the derivative recurrence

Differentiate the same generating function in x :

$$g_x = \frac{1}{2}(t - t^{-1})g, \quad g_x = \sum_{m=-\infty}^{\infty} J'_m(x)t^m. \quad (5.2.3)$$

The coefficient shifts are the same as before, with a minus sign between them:

$$2J'_m(x) = J_{m-1}(x) - J_{m+1}(x). \quad (5.2.4)$$

Now add and subtract (5.2.2) and (5.2.4):

$$2J_{m-1} = \frac{2m}{x}J_m + 2J'_m, \quad 2J_{m+1} = \frac{2m}{x}J_m - 2J'_m.$$

Solving each for J'_m gives, for $x \neq 0$,

$$J'_m = J_{m-1} - \frac{m}{x}J_m, \quad (5.2.5)$$

$$J'_m = \frac{m}{x}J_m - J_{m+1}. \quad (5.2.6)$$

These are exact symbolic identities. Repeated forward use at fixed x and large m can amplify roundoff; an identity need not be a stable numerical algorithm.

Consequences and low-order checks

Putting $m = 0$ in (5.2.6) gives

$$J_0'(x) = -J_1(x). \quad (5.2.7)$$

One series check fixes the sign:

$$\frac{d}{dx} \left(1 - \frac{x^2}{4} + \frac{x^4}{64} + \cdots \right) = -\frac{x}{2} + \frac{x^3}{16} + \cdots = -J_1(x).$$

The one-sided recurrences also turn products into derivatives. For $x \neq 0$,

$$\frac{d}{dx}(x^m J_m) = x^m \left(J_m' + \frac{m}{x} J_m \right) = x^m J_{m-1}, \quad (5.2.8)$$

$$\frac{d}{dx}(x^{-m} J_m) = x^{-m} \left(J_m' - \frac{m}{x} J_m \right) = -x^{-m} J_{m+1}. \quad (5.2.9)$$

The first equality on each line is the product rule; the second uses the corresponding one-sided derivative recurrence. These forms are useful in integrals and boundary conditions.

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Bessel's real integral representation

For real x and integer m , start from (5.1.8) and split the integral into $[0, \pi]$ and $[\pi, 2\pi]$. In the second half put $\theta = 2\pi - u$. Then $d\theta = -du$, with limits $u : \pi \rightarrow 0$, so reversing them removes the minus sign. Since $\sin(2\pi - u) = -\sin u$,

$$\begin{aligned}\int_{\pi}^{2\pi} e^{i(x \sin \theta - m\theta)} d\theta &= \int_0^{\pi} e^{-i(x \sin u - mu)} e^{-2\pi im} du \\ &= \int_0^{\pi} e^{-i(x \sin u - mu)} du.\end{aligned}$$

Here $e^{-2\pi im} = 1$ because m is an integer. Adding the first half gives

$$\int_0^{\pi} [e^{i(x \sin u - mu)} + e^{-i(x \sin u - mu)}] du = 2 \int_0^{\pi} \cos(mu - x \sin u) du.$$

Thus

$$J_m(x) = \frac{1}{\pi} \int_0^{\pi} \cos(m\theta - x \sin \theta) d\theta, \quad m \in \mathbb{Z}_{\geq 0}. \quad (5.3.1)$$

In particular,

$$J_0(x) = \frac{1}{\pi} \int_0^{\pi} \cos(x \sin \theta) d\theta. \quad (5.3.2)$$

The same derivation works for negative integer m , consistently with (5.1.6).

Contour form (Schl\"afli / Cauchy)

Laurent extraction (5.1.2) applied to the generating function gives

$$J_n(x) = \frac{1}{2\pi i} \oint_C e^{\frac{x}{2}(t-t^{-1})} t^{-n-1} dt, \quad n \in \mathbb{Z}, \quad (5.3.3)$$

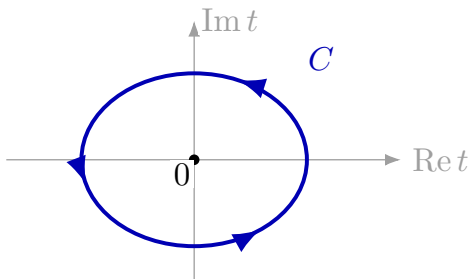
where C is a simple counterclockwise contour encircling zero once, without passing through zero. On the unit circle $t = e^{i\theta}$,

$$dt = it d\theta, \quad \frac{x}{2}(t - t^{-1}) = ix \sin \theta, \quad t^{-n-1} dt = ie^{-in\theta} d\theta.$$

Substitution recovers (5.1.8). This is the Chapter 1 coefficient formula applied to the whole Bessel family.

Contours in the Schlöfli representation

integer order



t^{-n-1} single-valued
one counterclockwise circuit

For integer n , t^{-n-1} is single-valued. The contour travels counterclockwise once around zero; its shape can change without changing the coefficient integral.

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Small x : leading term and first corrections

For fixed $m \geq 0$, take the first three terms of (5.1.3):

$$J_m(x) = \frac{(x/2)^m}{m!} - \frac{(x/2)^{m+2}}{(m+1)!} + \frac{(x/2)^{m+4}}{2!(m+2)!} + O(x^{m+6}).$$

Factor out $(x/2)^m/m!$. Using $(m+1)! = (m+1)m!$ and $(m+2)! = (m+2)(m+1)m!$ gives

$$J_m(x) = \frac{(x/2)^m}{m!} \left[1 - \frac{x^2}{4(m+1)} + \frac{x^4}{32(m+1)(m+2)} + O(x^6) \right]. \quad (5.4.1)$$

In particular,

$$J_m(x) = \frac{(x/2)^m}{\Gamma(m+1)} + O(x^{m+2}), \quad x \rightarrow 0, \quad (5.4.2)$$

with the order held fixed. The gamma notation agrees with the general-order series introduced earlier. Thus $J_0(0) = 1$ and $J_m(0) = 0$ for $m \geq 1$.

Large x : stationary phase on the unit circle

For fixed integer n , write the integral as

$$J_n(x) = \frac{1}{2\pi} \int_0^{2\pi} e^{-in\theta} e^{ix \sin \theta} d\theta.$$

When x is large, most neighboring contributions cancel through rapid oscillation. The main contributions come from points where the phase changes most slowly:

$$\frac{d}{d\theta} \sin \theta = \cos \theta = 0 \implies \theta_+ = \frac{\pi}{2}, \quad \theta_- = \frac{3\pi}{2}.$$

They are $t = i$ and $t = -i$ on the unit circle. We apply the stationary-phase rule of Chapter 1 by replacing the phase locally by its quadratic Taylor polynomial. This local calculation gives the leading terms; the stated remainder follows from the stationary-phase expansion for this smooth periodic integral.

Large x : stationary phase on the unit circle (cont.)

Near $\theta_+ = \pi/2$. Set $\theta = \pi/2 + u$. Then

$$\sin(\pi/2 + u) = 1 - \frac{1}{2}u^2 + O(u^4), \quad e^{-in\theta} = e^{-in\pi/2}(1 + O(u)).$$

Keeping the constant amplitude and quadratic phase gives the local contribution

$$J_n^{(+)}(x) \approx \frac{e^{ix - in\pi/2}}{2\pi} \int_{-\infty}^{\infty} e^{-ixu^2/2} du.$$

The substitution $v = \sqrt{x} u$ explains the factor $x^{-1/2}$. The oscillatory Gaussian, understood with a factor $e^{-\varepsilon u^2}$ followed by $\varepsilon \downarrow 0$, is

$$\int_{-\infty}^{\infty} e^{-ixu^2/2} du = e^{-i\pi/4} \sqrt{\frac{2\pi}{x}}. \quad (5.4.3)$$

Hence

$$J_n^{(+)}(x) = \frac{e^{i(x - n\pi/2 - \pi/4)}}{\sqrt{2\pi x}} + O(x^{-3/2}).$$

Large x : stationary phase on the unit circle (cont.)

Near $\theta_- = 3\pi/2$. Set $\theta = 3\pi/2 + u$. Now

$$\sin(3\pi/2 + u) = -1 + \frac{1}{2}u^2 + O(u^4), \quad e^{-i3n\pi/2} = e^{in\pi/2} \quad (n \in \mathbb{Z}).$$

Thus the quadratic phase has the opposite sign:

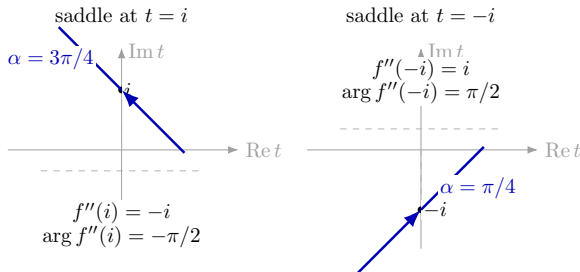
$$\int_{-\infty}^{\infty} e^{ixu^2/2} du = e^{i\pi/4} \sqrt{\frac{2\pi}{x}}. \quad (5.4.4)$$

The second contribution is

$$J_n^{(-)}(x) = \frac{e^{-i(x-n\pi/2-\pi/4)}}{\sqrt{2\pi x}} + O(x^{-3/2}).$$

The two terms are complex conjugates. Adding them will therefore produce a real cosine.

Steepest descent and the unit-circle saddles



The same points are complex saddles of $f(t) = \frac{1}{2}(t - t^{-1})$:

$$f'(t) = \frac{1}{2}(1 + t^{-2}) = 0 \implies t = \pm i, \quad f''(i) = -i, \quad f''(-i) = i.$$

Chapter 1's descent rule gives directions $3\pi/4$ at i and $\pi/4$ at $-i$, each modulo π . The unit-circle calculation already evaluates these contributions and fixes their phases; another contour calculation is unnecessary.

Large x : combining the two saddle contributions

Set $\chi_n(x) = x - n\pi/2 - \pi/4$. Adding the two local contributions gives

$$J_n(x) = \frac{e^{i\chi_n(x)} + e^{-i\chi_n(x)}}{\sqrt{2\pi x}} + O(x^{-3/2}).$$

Since $e^{i\chi} + e^{-i\chi} = 2 \cos \chi$,

$$J_n(x) = \sqrt{\frac{2}{\pi x}} \cos\left(x - \frac{n\pi}{2} - \frac{\pi}{4}\right) + O(x^{-3/2}), \quad x \rightarrow +\infty. \quad (5.4.5)$$

This holds for fixed integer n . We use an absolute error rather than a ratio asymptotic because the leading cosine has infinitely many zeros.

For fixed noninteger real order the same formula holds with n replaced by ν (quoted; DLMF §10.17). The integer Fourier integral does not derive that extension. In both cases the order is held fixed as $x \rightarrow +\infty$.

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Reduction of order: constructing a second solution

Dividing Bessel's equation by x^2 for $x > 0$ gives

$$y'' + \frac{1}{x}y' + \left(1 - \frac{m^2}{x^2}\right)y = 0. \quad (5.5.1)$$

Chapter 3, §3.3 derived the reduction-of-order formula $y_2 = y_1 \int e^{-\int P dx} / y_1^2 dx$ for $y'' + Py' + Qy = 0$. Here $P = 1/x$ and $e^{-\int P dx} = 1/x$, so

$$y_2(x) = J_m(x) \int_{x_*}^x \frac{ds}{sJ_m^2(s)}.$$

This formula is valid on an interval containing x_* where J_m has no zero. It supplies the general solution there:

$$y(x) = AJ_m(x) + BJ_m(x) \int_{x_*}^x \frac{ds}{sJ_m^2(s)}. \quad (5.5.2)$$

Reduction of order: constructing a second solution (cont.)

For $v = \int_{x_*}^x ds/[sJ_m^2(s)]$, the product-rule terms cancel:

$$W[J_m, vJ_m] = J_m(v'J_m + vJ_m') - J_m'(vJ_m) = v'J_m^2 = \frac{1}{x}.$$

The conventional Neumann function has normalization

$$W[J_m, Y_m] = \frac{2}{\pi x}. \quad (5.5.3)$$

Thus $y_2 = (\pi/2)Y_m + cJ_m$. Multiplying by $2/\pi$ fixes the Wronskian, but not the added multiple of J_m . The definitions below fix that remaining choice. The integral formula is local; the resulting solution extends across positive zeros of J_m .

Noninteger order: definition and independence of Y_ν

Bessel's equation depends on the order only through ν^2 . Therefore, if $J_\nu(x)$ is a solution, then $J_{-\nu}(x)$ is also a solution. For fixed real $\nu \notin \mathbb{Z}$, define

$$Y_\nu(x) = \frac{J_\nu(x) \cos(\pi\nu) - J_{-\nu}(x)}{\sin(\pi\nu)}. \quad (5.5.4)$$

To verify independence, first compute $W[J_\nu, J_{-\nu}]$ from their leading small- x terms. For $\nu \notin \mathbb{Z}$, retaining only the $k = 0$ term of the series (5.1.4) gives $(x/2)^{\pm\nu}/\Gamma(1 \pm \nu)$, so that

$$\begin{aligned} J_\nu(x) &\sim Ax^\nu, & A &= \frac{2^{-\nu}}{\Gamma(\nu+1)}, \\ J_{-\nu}(x) &\sim Bx^{-\nu}, & B &= \frac{2^\nu}{\Gamma(1-\nu)}. \end{aligned}$$

Thus

$$\begin{aligned} W[J_\nu, J_{-\nu}] &= J_\nu J'_{-\nu} - J'_\nu J_{-\nu} \\ &\sim (Ax^\nu)(-\nu Bx^{-\nu-1}) - (\nu Ax^{\nu-1})(Bx^{-\nu}) \\ &= -2\nu ABx^{-1}. \end{aligned}$$

Noninteger order: definition and independence of Y_ν (cont.)

Abel's identity from Chapter 3 gives $W' + W/x = 0$, hence $(xW)' = 0$ and $W = C_\nu/x$. Thus the leading calculation fixes the exact constant. The small- x limit above therefore fixes $C_\nu = -2\nu AB$. Now

$$\begin{aligned} AB &= \frac{1}{\Gamma(\nu+1)\Gamma(1-\nu)} \\ &= \frac{1}{\nu\Gamma(\nu)\Gamma(1-\nu)} \\ &= \frac{\sin(\pi\nu)}{\pi\nu}, \end{aligned}$$

where the last step uses Euler's reflection formula $\Gamma(\nu)\Gamma(1-\nu) = \pi/\sin(\pi\nu)$. Hence

$$W[J_\nu, J_{-\nu}] = -\frac{2\sin(\pi\nu)}{\pi x}. \quad (5.5.5)$$

Noninteger order: definition and independence of Y_ν (cont.)

Using linearity of the Wronskian in its second argument together with $W[J_\nu, J_\nu] = 0$, the Wronskian of (5.5.4) is therefore

$$\begin{aligned} W[J_\nu, Y_\nu] &= \frac{1}{\sin(\pi\nu)} W[J_\nu, J_\nu \cos(\pi\nu) - J_{-\nu}] \\ &= \frac{1}{\sin(\pi\nu)} [\cos(\pi\nu) W[J_\nu, J_\nu] - W[J_\nu, J_{-\nu}]] \\ &= -\frac{1}{\sin(\pi\nu)} W[J_\nu, J_{-\nu}] \\ &= \frac{2}{\pi x}, \end{aligned}$$

which establishes independence and the normalization $W[J_\nu, Y_\nu] = 2/(\pi x)$ for every noninteger ν .

At integer order the quotient has a removable $0/0$ form. Its continuous limit defines Y_m ; the next frame states that definition. The Wronskian also passes to the limit and remains $2/(\pi x)$, so independence is retained.

Integer order: define the second solution by a limit

At an integer m , $J_{-m} = (-1)^m J_m$, $\cos(\pi m) = (-1)^m$, and $\sin(\pi m) = 0$. Thus direct substitution into (5.5.4) gives $0/0$, not a value for Y_m .

The conventional definition is the continuous limit

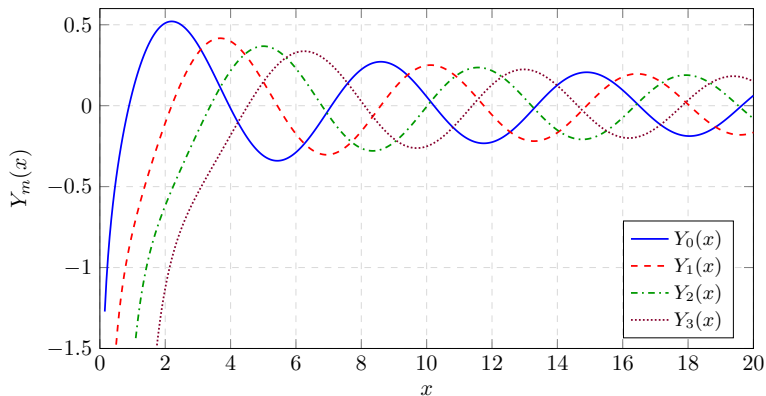
$$Y_m(x) = \lim_{\nu \rightarrow m} \frac{J_\nu(x) \cos(\pi \nu) - J_{-\nu}(x)}{\sin(\pi \nu)}, \quad x > 0. \quad (5.5.6)$$

We use the standard result that this limit, and its first two x -derivatives, exist. It therefore still solves the integer-order Bessel equation. The Wronskian passes to the limit:

$$W[J_m, Y_m] = \frac{2}{\pi x} \neq 0.$$

So Y_m is independent of J_m , even though J_{-m} is not. Unlike a Wronskian normalization alone, this limit also fixes the additive multiple of J_m . We use Y_m directly; order derivatives are not needed below.

Bessel functions of the second kind



The Neumann functions are independent of J_m and singular at $x = 0$. For large positive x , J_m and Y_m have the same $x^{-1/2}$ envelope and phases shifted by $\pi/2$.

Small- x behavior of Y_m

First let $\nu > 0$ be fixed and noninteger. From the first term of the series,

$$J_\nu(x) = \frac{1}{\Gamma(\nu+1)} \left(\frac{x}{2}\right)^\nu [1 + O(x^2)],$$
$$J_{-\nu}(x) = \frac{1}{\Gamma(1-\nu)} \left(\frac{x}{2}\right)^{-\nu} [1 + O(x^2)].$$

Their magnitudes have orders x^ν and $x^{-\nu}$, respectively. Since $x^\nu/x^{-\nu} = x^{2\nu} \rightarrow 0$, the $J_{-\nu}$ term dominates in (5.5.4). Therefore

$$Y_\nu(x) = -\frac{1}{\Gamma(1-\nu)\sin(\pi\nu)} \left(\frac{x}{2}\right)^{-\nu} [1 + o(1)].$$

Euler's reflection formula gives

$$\Gamma(\nu)\Gamma(1-\nu) = \frac{\pi}{\sin(\pi\nu)} \quad \implies \quad \frac{1}{\Gamma(1-\nu)\sin(\pi\nu)} = \frac{\Gamma(\nu)}{\pi}.$$

Consequently,

$$Y_\nu(x) = -\frac{\Gamma(\nu)}{\pi} \left(\frac{2}{x}\right)^\nu [1 + o(1)], \quad x \rightarrow 0^+, \quad (5.5.7)$$

for fixed positive noninteger ν .

Small- x behavior of Y_m (cont.)

For positive integer m , Chapter 3 obtained the small- x behavior directly from reduction of order, without interchanging limits in the order:

$$J_m(x) \int^x \frac{ds}{sJ_m^2(s)} \sim \frac{x^m}{2^m m!} \int^x 2^{2m} (m!)^2 s^{-2m-1} ds = -2^{m-1} (m-1)! x^{-m}.$$

Multiplication by the normalization factor $2/\pi$ gives

$$Y_m(x) = -\frac{\Gamma(m)}{\pi} \left(\frac{2}{x}\right)^m [1 + o(1)], \quad m = 1, 2, \dots, \quad x \rightarrow 0^+. \quad (5.5.8)$$

For $m = 0$, $J_0(x) = 1 + O(x^2)$, so the integrand is $1/[xJ_0^2(x)] = 1/x + O(x)$. Its integral is $\ln x + C + O(x^2)$. The order-limit definition fixes the constant and gives the standard formula

$$Y_0(x) = \frac{2}{\pi} \left[\ln \left(\frac{x}{2} \right) + \gamma \right] + O(x^2 |\ln x|), \quad x \rightarrow 0^+. \quad (5.5.9)$$

The leading logarithm follows from $\int dx/x$; the additive constant is quoted from the standard normalization (DLMF §10.8), not determined by that leading integral.

Large- x behavior of J_ν and Y_ν

For fixed real order ν , set

$$\chi_\nu(x) = x - \frac{\nu\pi}{2} - \frac{\pi}{4}, \quad A(x) = \sqrt{\frac{2}{\pi x}}.$$

The preceding large- x result and its quoted noninteger extension give

$$J_\nu(x) = A(x) \cos \chi_\nu(x) + O(x^{-3/2}). \quad (5.5.10)$$

Since $\chi_{-\nu} = \chi_\nu + \pi\nu$, insert the leading terms into the definition of Y_ν for fixed noninteger ν :

$$Y_\nu(x) = A(x) \frac{\cos \chi_\nu \cos(\pi\nu) - \cos(\chi_\nu + \pi\nu)}{\sin(\pi\nu)} + O(x^{-3/2}).$$

The identity $\cos(\chi + \beta) = \cos \chi \cos \beta - \sin \chi \sin \beta$ makes the numerator $\sin \chi_\nu \sin(\pi\nu)$, so

$$Y_\nu(x) = \sqrt{\frac{2}{\pi x}} \sin \chi_\nu(x) + O(x^{-3/2}), \quad x \rightarrow +\infty. \quad (5.5.11)$$

The same sine law holds at integer order (quoted; DLMF §10.17); the division above is not a proof of that limit. Thus J_m and Y_m have the same leading envelope and a phase difference of $\pi/2$.

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Hankel functions: Bessel functions of the third kind

Define

$$H_m^{(1)}(x) = J_m(x) + iY_m(x), \quad (5.6.1)$$

$$H_m^{(2)}(x) = J_m(x) - iY_m(x). \quad (5.6.2)$$

Let

$$\chi_m(x) = x - \frac{m\pi}{2} - \frac{\pi}{4}.$$

Using (5.5.10) and (5.5.11),

$$\begin{aligned} H_m^{(1)}(x) &= \sqrt{\frac{2}{\pi x}} (\cos \chi_m + i \sin \chi_m) + O(x^{-3/2}) \\ &= \sqrt{\frac{2}{\pi x}} e^{i\chi_m(x)} + O(x^{-3/2}), \end{aligned}$$

$$\begin{aligned} H_m^{(2)}(x) &= \sqrt{\frac{2}{\pi x}} (\cos \chi_m - i \sin \chi_m) + O(x^{-3/2}) \\ &= \sqrt{\frac{2}{\pi x}} e^{-i\chi_m(x)} + O(x^{-3/2}). \end{aligned}$$

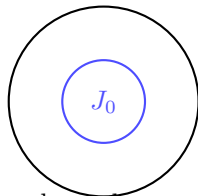
Hankel functions: Bessel functions of the third kind (cont.)

With $k > 0$, $\omega > 0$, and time dependence $e^{-i\omega t}$,

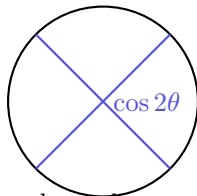
$$H_m^{(1)}(kr)e^{-i\omega t} \sim \sqrt{\frac{2}{\pi kr}} e^{i(kr - \omega t - m\pi/2 - \pi/4)}, \quad r \rightarrow \infty$$

for fixed m . Its constant-phase surfaces move toward larger r , so it is outgoing; $H_m^{(2)}$ is incoming. The Hankel functions are exact radial solutions for $r > 0$; only these leading exponential forms are far-field approximations. Both are singular at the origin, unlike the regular J_m modes.

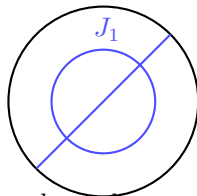
Discrete radial modes on the unit disk



angular order $m = 0$
radial mode 2



angular order $m = 2$
radial mode 1
 θ is the polar angle



angular order $m = 1$
radial mode 2

The boundary condition $J_m(\alpha_{m,j}) = 0$ places a radial zero at the clamped rim $r = 1$. Distinct zeros give distinct Sturm–Liouville eigenvalues $\lambda_j = \alpha_{m,j}^2$.

Scaling Bessel's equation and obtaining Sturm–Liouville form

Return to the disk problem in Chapter 4, §4.2. For fixed integer $m \geq 0$ and $y(s) = J_m(\alpha s)$, the chain rule gives $y' = \alpha J'_m$ and $y'' = \alpha^2 J''_m$. The rescaled equation is

$$s^2 y'' + sy' + (\alpha^2 s^2 - m^2)y = 0. \quad (5.6.3)$$

Divide by $s > 0$ and use $(sy')' = sy'' + y'$:

$$(sy')' - \frac{m^2}{s}y = -\alpha^2 sy. \quad (5.6.4)$$

Thus $p = s$, $q = -m^2/s$, $w = s$, and $\lambda = \alpha^2$. The weight gives

$$\langle u, v \rangle = \int_0^1 u(s) \overline{v(s)} s \, ds, \quad (5.6.5)$$

linear in the first slot, as in Chapter 4. The disk modes are regular at the origin and satisfy $y(1) = 0$ at the rim.

Scaling Bessel's equation and obtaining Sturm–Liouville form (cont.)

Check both endpoints. Green's identity leaves the boundary expression $s(uv' - u'v)$. For two regular Bessel solutions,

	u, v	u', v'	$s(uv' - u'v)$
$m \geq 1$	$O(s^m)$	$O(s^{m-1})$	$O(s^{2m})$
$m = 0$	$1 + O(s^2)$	$O(s)$	$O(s^2)$

so the expression tends to zero at the origin in both cases. Singular Y_m modes are excluded by the regularity condition. At the rim both mode values vanish. This is why Chapter 4's orthogonality result applies; $p(0) = 0$ alone would not suffice. The remaining task is new: compute the diagonal norm needed for expansion coefficients.

Fourier–Bessel orthogonality: distinct eigenvalues

For positive parameters a, b , let $y_a(x) = J_m(ax)$ and $y_b(x) = J_m(bx)$. Chapter 4's Green identity applied to (5.6.4) gives

$$\frac{d}{dx} [x(y_b y'_a - y_a y'_b)] + (a^2 - b^2)xy_a y_b = 0.$$

Integrating and using the origin estimate just checked,

$$(a^2 - b^2) \int_0^1 xy_a y_b dx = - [x(y_b y'_a - y_a y'_b)]_0^1.$$

At $x = 1$, $y_a = J_m(a)$ and $y'_a = aJ'_m(a)$; similarly for b . Hence

$$\int_0^1 xJ_m(ax)J_m(bx) dx = \frac{bJ_m(a)J'_m(b) - aJ_m(b)J'_m(a)}{a^2 - b^2}, \quad a \neq b. \quad (5.6.6)$$

The order of the two numerator terms comes from the minus sign before the boundary term.

Fourier–Bessel orthogonality: distinct eigenvalues (cont.)

Now choose distinct positive zeros $a = \alpha_{m,j}$ and $b = \alpha_{m,k}$ of J_m . Both $J_m(a)$ and $J_m(b)$ are zero, so the numerator vanishes while $a^2 - b^2 \neq 0$:

$$\int_0^1 x J_m(\alpha_{m,j} x) J_m(\alpha_{m,k} x) dx = 0, \quad j \neq k. \quad (5.6.7)$$

For the diagonal case $j = k$, both numerator and denominator vanish instead. We therefore take a limit, rather than substitute $a = b$ into (5.6.6).

Fourier–Bessel normalization: the diagonal case

Existence of the zeros. For fixed integer $m \geq 0$, the large- x cosine law (5.4.5) oscillates with amplitude $\sim x^{-1/2}$ and phase advancing like x , so J_m changes sign infinitely often as $x \rightarrow +\infty$. Hence J_m has infinitely many positive zeros

$\alpha_{m,1} < \alpha_{m,2} < \cdots$. We write $\alpha_{m,j}$ for the j th.

Every positive zero a is simple: at $J_m(a) = 0$, the Wronskian gives

$-J'_m(a)Y_m(a) = 2/(\pi a) \neq 0$, so $J'_m(a) \neq 0$. This ensures that the norm in the coefficient formula is nonzero.

To compute the norm at such a zero, take the limit $b \rightarrow a$ in (5.6.6). The integral passes to the limit by continuity of the integrand on the compact interval $0 \leq x \leq 1$.

Write the numerator and denominator as functions of b with a fixed and $J_m(a) = 0$:

$$N(b) := bJ_m(a)J'_m(b) - aJ_m(b)J'_m(a) = -aJ_m(b)J'_m(a),$$

$$D(b) := a^2 - b^2.$$

Fourier–Bessel normalization: the diagonal case (cont.)

Both $N(a)$ and $D(a)$ vanish, so the limit is a $0/0$ form. Apply L'Hôpital: differentiate with respect to b ,

$$\begin{aligned}N'(b) &= -aJ'_m(b)J'_m(a), \\D'(b) &= -2b.\end{aligned}$$

Therefore

$$\lim_{b \rightarrow a} \frac{N(b)}{D(b)} = \lim_{b \rightarrow a} \frac{N'(b)}{D'(b)} = \frac{-aJ'_m(a)J'_m(a)}{-2a} = \frac{1}{2} [J'_m(a)]^2,$$

where $a > 0$. Therefore

$$\int_0^1 x J_m^2(ax) dx = \frac{1}{2} [J'_m(a)]^2 \quad \text{at a positive zero } a \text{ of } J_m. \quad (5.6.8)$$

At a zero, $J_m(a) = 0$, so the derivative recurrence gives

$$J'_m(a) = \frac{m}{a} J_m(a) - J_{m+1}(a) = -J_{m+1}(a).$$

Squaring removes the sign. Combining the diagonal and off-diagonal cases gives

$$\int_0^1 J_m(\alpha_{m,j}x) J_m(\alpha_{m,k}x) x dx = \frac{1}{2} J_{m+1}^2(\alpha_{m,j}) \delta_{jk}. \quad (5.6.9)$$

Fourier–Bessel coefficients

For a radial function of fixed angular order m , write

$$f(x) = \sum_{j=1}^{\infty} a_j J_m(\alpha_{m,j}x), \quad 0 < x < 1. \quad (5.6.10)$$

As for the sine series in Chapter 4, equality means convergence in the weighted mean-square norm $\|g\|^2 = \int_0^1 |g(x)|^2 x \, dx$.

Apply the same projection formula: multiply by the k th mode and the weight x , then integrate. Orthogonality removes every term except $j = k$:

$$\int_0^1 f(x) J_m(\alpha_{m,k}x) x \, dx = a_k \int_0^1 J_m^2(\alpha_{m,k}x) x \, dx = \frac{a_k}{2} J_{m+1}^2(\alpha_{m,k}).$$

Solving gives

$$a_k = \frac{2}{J_{m+1}^2(\alpha_{m,k})} \int_0^1 f(r) J_m(\alpha_{m,k}r) r \, dr. \quad (5.6.11)$$

The denominator is the norm just derived, not an arbitrary normalization.

These disk modes are complete in the weighted mean-square space (a quoted Sturm–Liouville result). Finite weighted norm therefore suffices for the expansion, but not for pointwise equality.

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Formula sheet: J_m

For integer $m \geq 0$,

$$x^2 J_m''(x) + x J_m'(x) + (x^2 - m^2) J_m(x) = 0,$$

$$J_m(x) = \sum_{k=0}^{\infty} \frac{(-1)^k}{k!(m+k)!} \left(\frac{x}{2}\right)^{2k+m},$$

$$J_{-m}(x) = (-1)^m J_m(x).$$

The generating function, valid for $t \neq 0$, is

$$e^{\frac{x}{2}(t-t^{-1})} = \sum_{m=-\infty}^{\infty} J_m(x) t^m.$$

Formula sheet: J_m (cont.)

The recurrences are

$$J_{m-1}(x) + J_{m+1}(x) = \frac{2m}{x} J_m(x), \quad x \neq 0,$$

$$2J'_m(x) = J_{m-1}(x) - J_{m+1}(x),$$

$$J'_m(x) = J_{m-1}(x) - \frac{m}{x} J_m(x), \quad x \neq 0,$$

$$J'_m(x) = \frac{m}{x} J_m(x) - J_{m+1}(x), \quad x \neq 0.$$

The preferred integral representations are

$$J_m(x) = \frac{1}{\pi} \int_0^\pi \cos(m\theta - x \sin \theta) d\theta, \quad m \geq 0,$$

$$J_n(x) = \frac{1}{2\pi} \int_0^{2\pi} e^{i(x \sin \theta - n\theta)} d\theta, \quad n \in \mathbb{Z},$$

with the optional contour form

$$J_n(x) = \frac{1}{2\pi i} \oint_C e^{\frac{x}{2}(t-t^{-1})} t^{-n-1} dt, \quad n \in \mathbb{Z}.$$

Formula sheet: J_m (cont.)

For fixed m ,

$$J_m(x) = \frac{1}{m!} \left(\frac{x}{2}\right)^m \left[1 - \frac{x^2}{4(m+1)} + O(x^4)\right], \quad x \rightarrow 0,$$

$$J_m(x) = \sqrt{\frac{2}{\pi x}} \cos\left(x - \frac{m\pi}{2} - \frac{\pi}{4}\right) + O(x^{-3/2}), \quad x \rightarrow +\infty.$$

Formula sheet: Y_m , Hankel, Fourier–Bessel

For noninteger ν ,

$$Y_\nu(x) = \frac{J_\nu(x) \cos(\pi\nu) - J_{-\nu}(x)}{\sin(\pi\nu)},$$

and for integer m ,

$$Y_m(x) = \lim_{\nu \rightarrow m} \frac{J_\nu(x) \cos(\pi\nu) - J_{-\nu}(x)}{\sin(\pi\nu)}, \quad x > 0.$$

The normalization is

$$W[J_m, Y_m] = \frac{2}{\pi x}.$$

For fixed integer $m \geq 1$,

$$Y_m(x) = -\frac{\Gamma(m)}{\pi} \left(\frac{2}{x}\right)^m [1 + o(1)], \quad x \rightarrow 0^+,$$

whereas

$$Y_0(x) = \frac{2}{\pi} \left[\log\left(\frac{x}{2}\right) + \gamma \right] + O(x^2 |\log x|).$$

For fixed m and $x \rightarrow +\infty$,

$$Y_m(x) = \sqrt{\frac{2}{\pi x}} \sin\left(x - \frac{m\pi}{2} - \frac{\pi}{4}\right) + O(x^{-3/2}).$$

Formula sheet: Y_m , Hankel, Fourier–Bessel (cont.)

The Hankel functions are

$$\begin{aligned}H_m^{(1)} &= J_m + iY_m, & H_m^{(1)}(x) &= \sqrt{\frac{2}{\pi x}} e^{i\chi_m(x)} + O(x^{-3/2}), \\H_m^{(2)} &= J_m - iY_m, & H_m^{(2)}(x) &= \sqrt{\frac{2}{\pi x}} e^{-i\chi_m(x)} + O(x^{-3/2}),\end{aligned}$$

with $\chi_m(x) = x - m\pi/2 - \pi/4$.

For the positive zeros $\alpha_{m,j}$ of J_m ,

$$\int_0^1 J_m(\alpha_{m,j}x) J_m(\alpha_{m,k}x) x \, dx = \frac{1}{2} J_{m+1}^2(\alpha_{m,j}) \delta_{jk},$$

If a radial function of angular order m is expanded on the disk as

$$f(r) = \sum_{j=1}^{\infty} a_j J_m(\alpha_{m,j}r), \quad 0 < r < 1,$$

Formula sheet: Y_m , Hankel, Fourier–Bessel (cont.)

then multiplying by $J_m(\alpha_{m,k}r)r$, integrating over $(0, 1)$ and using the orthogonality relation above (only the term $j = k$ survives) gives the coefficients

$$a_j = \frac{2}{J_{m+1}^2(\alpha_{m,j})} \int_0^1 f(r) J_m(\alpha_{m,j}r) r \, dr,$$

as derived in (5.6.10)–(5.6.11).